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Skewness and Concentration in Nordic Equity Markets: Implications for Portfolio Construction

Working on it

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Abstract

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1 Introduction

Research conducted in U.S. equity markets shows that the majority of buy-and-hold common stock returns are outperformed by returns on one-month U.S. Treasury bills held over the same investment horizon (Bessembinder, 2018). This finding challenges the conventional view of equity markets as broadly rewarding for long-term investors and suggests that aggregate market returns may obscure substantial heterogeneity in individual stock outcomes.

Equity markets play a central role in long-term wealth accumulation, pension systems, and capital allocation. Market performance is therefore often summarized using long-run average returns or broad market indices. While such measures provide a useful aggregate perspective, they offer limited insight into how returns are generated at the individual stock level. Aggregate market returns in the U.S. mask a highly uneven distribution of outcomes across firms, where a small number of stocks generate large positive returns while the majority deliver modest or negative performance over long horizons (Bessembinder, 2018).

Previous research has documented this pattern extensively for the U.S. equity market. In particular, Professor Hendrik Bessembinder shows in his research (Bessembinder, 2018) that most individual stocks underperform certain risk-free assets over their lifetimes, while nearly all long-term net wealth creation is attributable to a small fraction of firms. These results imply that individual stock returns are highly right-skewed and that aggregate market performance is driven by a limited number of extreme winners.

However, comparable evidence for Nordic equity markets is limited. This is notable given that Nordic markets differ from the U.S. in terms of market size, sector composition, ownership structures and institutional characteristics. It is therefore of interest to examine whether a similar return structure is present in Nordic equity markets or whether these structural differences are associated with different patterns of return skewness and concentration.

Understanding the distribution of individual stock returns has important implications for portfolio construction and active asset management. In the presence of highly skewed return distributions, diversification primarily serves to increase the probability of capturing extreme positive outcomes rather than merely reducing volatility (Reinholtz, Fernbach, and Langhe, 2016). Similarly, the performance of concentrated portfolios and stock-picking strategies

must be assessed in a setting where underperformance relative to the market is the most likely outcome, despite the existence of a small number of highly successful investments.

By combining an empirical analysis of individual stock return distributions, a wealth-creation framework, and a portfolio-level evaluation, this study contributes to the literature by extending existing evidence beyond the U.S. market and by providing insights into the consequences of skewed return distributions for long-term investors in Nordic equity markets.

1.1 Kvartil

This thesis is conducted in collaboration with Kvartil, an asset management firm founded in 2018 that combines data, statistical analysis and systematic rules in its investment process to enhance decision-making and portfolio construction. The team at Kvartil has contributed with domain expertise, valuable insights and guidance throughout this thesis.

1.2 Purpose

The purpose of this study is to examine how different portfolio construction methods perform in a market environment characterized by positively skewed and concentrated individual stock returns.

1.3 Research questions

1. Do pooled return distributions in Nordic equity markets display significant positive skewness across different investment horizons?
2. How concentrated is long-term wealth creation across firms in the Nordic equity markets?
3. How does portfolio performance differ when stocks are sorted by skewness within size segments and combined at different levels of diversification?

1.4 Delimitations

To maintain a clear and well-defined scope, and to ensure consistency and availability of the data, this thesis applies several delimitations to the study design.

Nordic markets
This study is exclusively focused on the Nordic equity markets. By limiting the scope to the Nordic markets, the analysis aims to provide a more internally consistent setting in which observed return patterns can be attributed to underlying market dynamics rather than cross-market structural differences. The following markets are the ones considered in this study:

- Sweden
  - Nasdaq Stockholm Stock Exchange
  - Spotlight Stock Market
- Denmark
  - Nasdaq Copenhagen Stock Exchange

- Norway
  - Euronext Oslo Stock Exchange
- Finland
  - Nasdaq Helsinki Stock Exchange
- International
  - Nordic Growth Market

Time horizon and return periods
The analysis considers one-year, five-year, and lifetime return horizons in order to capture both short- to long-term return dynamics and long-run wealth creation. For the one-year and five-year horizons, the sample is restricted to the most recent 35-year period. This choice reflects a trade-off between obtaining a sufficiently large number of observations and maintaining relevance to the current and recent market environment.

Data selection
For the one-year and five-year return analyses, the dataset includes return observations from periods during which companies were publicly listed and actively traded on the Nordic equity markets. In contrast, the lifetime return analysis includes all companies that have been listed on the Nordic markets at some point in the sample time, covering their entire listing history from initial listing to delisting or the end of the sample period.

2 Background

This chapter provides the background necessary to understand the theoretical aspects of stock returns and their implications for investment outcomes. It begins by introducing the concept of return distributions and the statistical characteristics commonly observed in financial data. Particular attention is given to skewness, which reflects the asymmetric nature of stock returns and highlights the presence of extreme outcomes. The chapter then examines how such asymmetries translate into concentration in long-term market performance, where a limited number of firms account for a disproportionate share of total wealth creation. The concept of wealth creation is introduced as a framework for quantifying the economic contribution of individual firms to aggregate market outcomes. Finally, the chapter discusses the implications of the empirically observed skewness in pooled return observations, together with the concentration of long-term wealth creation, for portfolio construction. This provides the conceptual foundation for the analysis that follows, which investigates how different portfolio construction approaches perform in an environment characterized by asymmetric and highly concentrated return outcomes.

2.1 Return distribution on financial markets

In financial markets, asset returns are commonly analyzed through their statistical distributions. A return distribution describes how frequently different return outcomes occur over a given horizon and provides a compact representation of both typical and extreme outcomes. Understanding the shape of return distributions is therefore a fundamental prerequisite for analyzing risk, performance, and investment outcomes.

In empirical studies, returns are typically measured as arithmetic or logarithmic price changes over fixed intervals, such as daily, monthly, or annual horizons (Miskolczi, 2017). The collection of these observations over time forms an empirical return distribution for a specific firm, which can be summarized using descriptive statistics such as the mean, median, skewness, and standard deviation. While these summary measures provide useful information, they do not fully characterize the distributional shape of returns.

In much of classical financial theory, asset prices are commonly modeled as following a geometric Brownian motion. Under this assumption, continuously compounded logarithmic

returns are normally distributed, while prices themselves follow a lognormal process (Mota, 2012). However, a large amount of empirical evidence suggests that observed logarithmic return distributions frequently deviate from normality. Observed return distributions often exhibit features such as asymmetry, heavy tails, and excess kurtosis, meaning that extreme outcomes occur more frequently than predicted by a normal distribution (Cont, 2001). These deviations are observed across asset classes, markets, and time periods, and become particularly pronounced when returns are measured over longer horizons or at the level of individual stocks.

As a result, empirical return distributions are typically analyzed using non-parametric tools, such as histograms, which allows one to visualize the data without imposing strong distributional assumptions. These tools provide a flexible way to visualize and summarize return behavior and serve as a natural starting point for further distributional analysis.

This study adopts such an empirical perspective when examining stock returns in Nordic equity markets. Rather than assuming a particular parametric form, the analysis begins by documenting the observed distributional properties of returns across different horizons. This approach allows for a transparent assessment of how closely empirical return distributions align with standard theoretical assumptions and provides the foundation for subsequent analysis of asymmetry, concentration, and portfolio-level implications.

2.2 Skewness in return distribution

Skewness is a fundamental property of return distributions that captures the degree of asymmetry around the central tendency. In a positively skewed distribution, the right tail is longer or heavier than the left tail, implying that a small number of observations take on exceptionally large positive values, while the majority of observations are clustered around lower or moderate outcomes. (Doane and Seward, 2011)

In the context of financial returns, when individual stocks are considered as separate observations, their long-term return outcomes exhibit a highly positively skewed cross-sectional distribution. Empirically, the distribution of individual U.S stock returns is characterized by many small losses or modest gains, combined with a small number of extreme winners (Bessembinder, 2018). As a result, the distribution of returns is asymmetric, and measures of central tendency such as the mean and the median can differ substantially. The distinction between mean and median outcomes is economically important. In a positively skewed distribution, the mean is pulled upward by rare but extreme positive outcomes, while the median reflects the outcome experienced by a typical observation (Doane and Seward, 2011). This feature implies that overall market performance may be driven by a limited number of exceptional firms rather than reflecting the typical outcome across stocks.

Recent empirical evidence highlights the extent to which individual stock returns exhibit such skewed behavior. Bessembinder (2018) documents that, over long horizons, the majority of U.S. stocks generate returns below those of one month Treasury bills, while a relatively small subset of firms accounts for nearly all positive net wealth creation in the market. These findings illustrate that aggregate market performance is driven by a small number of extreme winners, while the typical stock contributes little to long-term wealth creation.

Importantly, skewness is a defining feature of equity return distributions with direct economic implications. When returns are highly skewed, investment outcomes become sensitive to the inclusion or exclusion of a small number of extreme winners. This property challenges

intuitive interpretations based on average performance and motivates a closer examination of how skewed stock-level returns aggregate at the market and portfolio level.

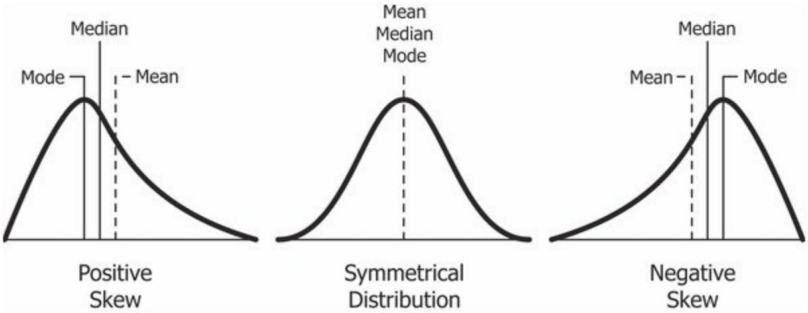

Figure 2.1: Illustration of different skewed distributions.

2.3 Long-term stock return and concentration

While skewness describes asymmetry in return distributions, it does not fully capture the economic importance of extreme outcomes. To assess their market-level relevance, it is necessary to examine how long-term stock returns and wealth creation are distributed across firms. When returns are measured over long horizons, such as several years or over a firm's entire publicly traded lifetime, each stock contributes a single cumulative outcome. Empirical evidence shows that this distribution is highly uneven, where most stocks generate modest or negative long-term returns, while a small number of firms achieve exceptionally large gains (Bessembinder, 2018).

This uneven contribution is captured by the concept of concentration, which refers to how total market wealth creation is distributed across firms. Aggregate market performance may therefore be driven by a limited subset of companies, even when the overall market exhibits strong long-term returns. Bessembinder (2018) formalizes this idea using the concept of wealth creation, defined as the cumulative value generated by a firm's equity in excess of a risk-free investment over its lifetime. Using this measure, he documents that nearly all net wealth creation in the U.S. equity market can be attributed to a small fraction of firms, while the typical stock underperforms risk-free assets (Bessembinder, 2018).

Wealth creation thus provides a direct link between skewness at the individual stock level and concentration in aggregate market outcomes. This perspective motivates an examination of whether similar patterns of concentration are present in Nordic equity markets and sets the stage for analyzing their implications for portfolio-level outcomes.

2.4 Implications for portfolio management

The presence of skewness and concentration in the return distribution have an impact when it comes to portfolio construction. When the long-term returns are unevenly distributed across firms, aggregate market performance is driven by a relatively small number of extreme positive outcomes. As a result, investment outcomes depend critically on whether a portfolio includes the firms responsible for the majority of long-term value creation.

In such an environment, the interpretation of average returns becomes less informative for individual investors. While the mean return across stocks may be positive, due to the positive skewness in the return distribution, the typical stock may deliver substantially lower

performance. This implies that investors holding only a limited number of securities face a significant probability of underperforming the broader market, even over long investment horizons. These characteristics highlight the role of diversification as a mechanism for increasing the likelihood of capturing the small subset of firms that generate the majority of long-term gains. Diversification, rather than solely reducing volatility (Reinholtz, Fernbach, and Langhe, 2016), improves the probability that a portfolio includes the extreme positive outcomes that drive aggregate market performance.

At the same time, different portfolio construction approaches may result in different exposure to these extreme outcomes. By sorting firms within size segments (small, mid and large cap) according to certain construction rules, and by varying the number of stocks included in each portfolio, the skewness exposure and diversification are analyzed to observe how they jointly shape portfolio return distributions in an environment characterized by concentrated wealth creation and an empirically skewed pooled distribution. This motivates an examination of how different portfolio construction rules affect the distribution of portfolio returns.

3 Methodology

This chapter outlines the methodological framework of the study. The objective is to examine whether Nordic equity markets exhibit skewed return distributions and concentrated wealth creation, and to analyze how these structural characteristics affect portfolio outcomes under different construction rules. The chapter begins with a review of relevant literature and a description of the data collection and processing procedures. It then presents the modeling framework, where skewness and wealth concentration are documented and portfolios are constructed based on predefined selection rules. Finally, an evaluation framework is introduced to assess differences between rule-based portfolios and to benchmark their performance against randomly formed portfolios. Together, these components provide a structured approach for linking stock-level return characteristics to portfolio-level performance in an environment characterized by asymmetric and concentrated return outcomes. All empirical analysis is conducted using data from LSEG Workspace and implemented in Python.

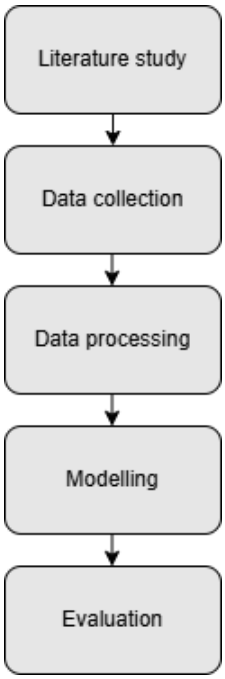

Figure 3.1: Overview of the methodology of the study.

3.1 Literature study

This study builds upon prior empirical research examining the distributional properties of individual stock returns and their implications for aggregate market outcomes. A contribution in this area is a study conducted by Bessembinder (2018), who analyzes the distribution of long-term stock returns using comprehensive data on U.S. equities. His research provides a framework for quantifying concentration in long-term equity returns and serves as a key reference point for this study. In addition to Bessembinders work, other research papers have been used that document that individual stock returns exhibit highly asymmetric distributions, characterized by positive skewness and the presence of extreme positive outcomes. This study builds on these approaches by applying similar distributional analyses to Nordic equity markets, specifically examining the empirical distribution of stock returns, quantify wealth creation and concentration, and construct rule-based portfolios to analyze how portfolio size and selection rules influence the distribution of portfolio outcomes.

3.2 Data collection

All necessary data for this study are obtained from LSEG Workspace.

3.2.1 Markets

To examine whether return skewness is present in Nordic equity markets, the analysis is based on firms listed on the major Nordic exchanges. Rather than focusing on a single country, the study covers multiple markets in order to capture the overall structure of return distributions in the Nordic region and to reduce the influence of country-specific effects.

The collected dataset includes all companies that were listed in any of the selected Nordic equity markets at some point during the sample period. This implies that the sample is defined by listing location rather than country of incorporation. Consequently, a subset of

the firms in the study may be registered or headquartered in another country while being publicly traded on a Nordic exchange.

From an investment perspective, what matters is where the security is traded and accessible to market participants, not necessarily where the firm is legally incorporated. By taking this approach the results describe the structure of returns in Nordic exchanges, rather than the performance of purely Nordic-domiciled firms. This distinction is important for interpretation, particularly when comparing findings to studies based on domestic firm samples in other regions.

The following markets are the ones included in the study:

- Sweden
  - Nasdaq Stockholm Stock Exchange
  - Spotlight Stock Market
- Denmark
  - Nasdaq Copenhagen Stock Exchange
- Norway
  - Euronext Oslo Stock Exchange
- Finland
  - Nasdaq Helsinki Stock Exchange
- International
  - Nordic Growth Market

3.2.2 Returns

This study is based on returns generated by publicly traded companies listed on the Nordic exchanges described in Section 3.2.1. For firms that were delisted during the sample period, the reason for delisting is identified and treated according to the defined data processing procedure 3.3.

In empirical finance and financial econometrics, returns are commonly expressed as logarithmic (continuously compounded) returns. This convention stems from their analytical convenience, particularly their time-additive property and their compatibility with geometric averaging and continuous compounding (Campbell, Lo, and MacKinlay, 1997). Accordingly, and in line with established practice, this study employs logarithmic returns throughout the empirical analysis.

3.3 Data processing

To enable a consistent and reliable empirical analysis, the collected data were subject to a structured process of cleaning, processing, and formatting. The data preparation was conducted in Excel, including the use of built-in functions and VBA scripts to manage large datasets and automate repetitive procedures.

A central component of the data processing concerned the treatment of firms that were

delisted during the sample period. For each delisted company, the reason for delisting was identified in order to ensure an economically correct measurement of the final return. In cases of bankruptcy or liquidation, the terminal return was recorded as a total loss, reflecting the fact that shareholders effectively lost their invested capital. In contrast, when a firm was delisted due to an acquisition or merger, the final observed return was calculated based on the last available price, including any compensation received by shareholders.

This distinction is crucial, as improper treatment of delistings may lead to survivorship bias (Financial Edge Training, 2022) in the distribution of returns, particularly in studies examining skewness and the concentration of long-term wealth creation.

3.4 Modeling

The modeling framework links stock-level return characteristics to portfolio-level outcomes. Specifically, the analysis proceeds in two stages. First, skewness and wealth concentration in individual stock returns are documented in order to establish the structural properties of the Nordic equity market. Second, portfolios are constructed based on predefined selection rules to examine how these structural features translate into portfolio performance.

3.4.1 Observation of skewness and concentration among returns

The first step of this study is to conduct an empirical observation of the distributional properties of stock returns in the Nordic market and their implications for wealth creation. This empirical observation can be divided into two stages.

First, the distribution of stock return observations is examined across different investment horizons, including one-year, five-year, and lifetime returns. The analysis focuses on the empirical shape of these distributions using graphical methods such as histograms. By documenting asymmetry and tail behavior without imposing parametric assumptions, this step establishes whether return outcomes exhibit pronounced skewness and uneven dispersion.

Second, firm-level wealth creation is evaluated to assess the concentration of long-term wealth creation within the Nordic market. Wealth creation is defined as the cumulative value generated by a firm's change in market capitalization in excess of a risk-free investment over its publicly traded lifetime. Firms are then compared based on their relative contribution to total market wealth creation, allowing for an assessment of how unevenly wealth creation is distributed across firms.

3.4.2 Portfolio construction

To examine the implications of skewness in individual stock returns, a range of portfolios is constructed and analyzed. The objective is to evaluate differences in performance across portfolios and to examine how their return distributions vary depending on underlying stock characteristics.

HÄR KOMMER VALET AV PORTFÖLJER ATT BESKRIVAS. SKEWNESS FÖR INDIVIDUELLA FÖRETAG, DIVERSIFIERNING OCH STORLEK (CAP) KOMMER ATT VARA PARAMETRAR FÖR DETTA. VIDARE DISKUSSION MED KVARTIL FÖRS FÖR ATT TA ÖNSKEMÅL FÖR FLER PARAMETRAR/REGLER I BEAKTNING.

3.5 Evaluation

This section outlines the framework used to evaluate the performance of the constructed portfolios. The evaluation is conducted in two complementary stages. First, rule-based portfolios are compared statistically to assess whether systematic differences in construction rules translate into differences in average returns. Second, each rule-based portfolio is evaluated relative to a benchmark distribution of randomly formed portfolios in order to assess whether its performance is distinguishable from random selection.

3.5.1 Statistical comparison of rule-based portfolios

To compare portfolios constructed under different rules, daily return differences between two portfolios are computed over the evaluation period, which is a period of the most recent five years. The mean return difference is then tested for statistical significance.

Because daily financial returns may exhibit heteroskedasticity and serial dependence, inference is conducted using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey and West, 1987). This approach provides robust standard errors that remain valid under general forms of time-series dependence. Statistical significance is assessed using the corresponding t-statistic and p-values derived from the asymptotic distribution of the estimator. The sign of the estimated mean difference determines which portfolio exhibits higher average performance, while statistical significance indicates whether the observed difference is unlikely to be driven by sampling variation.

3.5.2 Benchmarking against randomly formed portfolios

In addition to pairwise statistical tests, each rule-based portfolio is evaluated relative to a benchmark distribution of randomly formed portfolios. For each portfolio configuration, defined by size segment, diversification level, and (ANDRA REGLER), a large number of portfolios are generated through random selection of stocks under identical constraints.

This Monte Carlo simulation approach produces an empirical distribution of portfolio outcomes that reflects the range of performance attainable through random diversification within the same investment universe. The realized performance of the rule-based portfolio is then evaluated relative to this distribution, for example by assessing its percentile rank or by computing the proportion of randomly formed portfolios that achieve higher performance. By combining statistical inference between rule-based portfolios with simulation-based benchmarking against random portfolios, the evaluation framework provides a comprehensive assessment of whether systematic portfolio construction rules meaningfully alter investment outcomes in a market characterized by skewed and concentrated stock returns.

4 Theory

The main purpose of this chapter is to make it obvious for the reader that the report authors have made an effort to read up on related research and other information of relevance for the research questions. It is a question of trust. Can I as a reader rely on what the authors are saying? If it is obvious that the authors know the topic area well and clearly present their lessons learned, it raises the perceived quality of the entire report.

After having read the theory chapter it shall be obvious for the reader that the research questions are both well formulated and relevant.

The chapter must contain theory of use for the intended study, both in terms of technique and method. If a final thesis project is about the development of a new search engine for a certain application domain, the theory must bring up related work on search algorithms and related techniques, but also methods for evaluating search engines, including performance measures such as precision, accuracy and recall.

The chapter shall be structured thematically, not per author. A good approach to making a review of scientific literature is to use *Google Scholar* (which also has the useful function *Cite*). By iterating between searching for articles and reading abstracts to find new terms to guide further searches, it is fairly straight forward to locate good and relevant information, such as Walker, 1999.

Having found a relevant article one can use the function for viewing other articles that have cited this particular article, and also go through the article's own reference list. Among these articles on can often find other interesting articles and thus proceed further.

It can also be a good idea to consider which sources seem most relevant for the problem area at hand. Are there any special conference or journal that often occurs one can search in more detail in lists of published articles from these venues in particular. One can also search for the web sites of important authors and investigate what they have published in general.

This chapter is called either *Theory, Related Work*, or *Related Research*. Check with your supervisor.

5 Method

In this chapter, the method is described in a way which shows how the work was actually carried out. The description must be precise and well thought through. Consider the scientific term replicability. Replicability means that someone reading a scientific report should be able to follow the method description and then carry out the same study and check whether the results obtained are similar. Achieving replicability is not always relevant, but precision and clarity is.

Sometimes the work is separated into different parts, e.g. pre-study, implementation and evaluation. In such cases it is recommended that the method chapter is structured accordingly with suitable named sub-headings.

6 Results

This chapter presents the results. Note that the results are presented factually, striving for objectivity as far as possible. The results shall not be analyzed, discussed or evaluated. This is left for the discussion chapter.

In case the method chapter has been divided into subheadings such as pre-study, implementation and evaluation, the result chapter should have the same sub-headings. This gives a clear structure and makes the chapter easier to write.

In case results are presented from a process (e.g. an implementation process), the main decisions made during the process must be clearly presented and justified. Normally, alternative attempts, etc, have already been described in the theory chapter, making it possible to refer to it as part of the justification.

7 Discussion

This chapter contains the following sub-headings.

7.1 Results

Are there anything in the results that stand out and need be analyzed and commented on? How do the results relate to the material covered in the theory chapter? What does the theory imply about the meaning of the results? For example, what does it mean that a certain system got a certain numeric value in a usability evaluation; how good or bad is it? Is there something in the results that is unexpected based on the literature review, or is everything as one would theoretically expect?

7.2 Method

This is where the applied method is discussed and criticized. Taking a self-critical stance to the method used is an important part of the scientific approach.

A study is rarely perfect. There are almost always things one could have done differently if the study could be repeated or with extra resources. Go through the most important limitations with your method and discuss potential consequences for the results. Connect back to the method theory presented in the theory chapter. Refer explicitly to relevant sources.

The discussion shall also demonstrate an awareness of methodological concepts such as replicability, reliability, and validity. The concept of replicability has already been discussed in the Method chapter (5). Reliability is a term for whether one can expect to get the same results if a study is repeated with the same method. A study with a high degree of reliability has a large probability of leading to similar results if repeated. The concept of validity is, somewhat simplified, concerned with whether a performed measurement actually measures what one thinks is being measured. A study with a high degree of validity thus has a high level of credibility. A discussion of these concepts must be transferred to the actual context of the study.

The method discussion shall also contain a paragraph of source criticism. This is where the authors' point of view on the use and selection of sources is described.

In certain contexts it may be the case that the most relevant information for the study is not to be found in scientific literature but rather with individual software developers and open

source projects. It must then be clearly stated that efforts have been made to gain access to this information, e.g. by direct communication with developers and/or through discussion forums, etc. Efforts must also be made to indicate the lack of relevant research literature. The precise manner of such investigations must be clearly specified in a method section. The paragraph on source criticism must critically discuss these approaches.

Usually however, there are always relevant related research. If not about the actual research questions, there is certainly important information about the domain under study.

7.3 The work in a wider context

There must be a section discussing ethical and societal aspects related to the work. This is important for the authors to demonstrate a professional maturity and also for achieving the education goals. If the work, for some reason, completely lacks a connection to ethical or societal aspects this must be explicitly stated and justified in the section Delimitations in the introduction chapter.

In the discussion chapter, one must explicitly refer to sources relevant to the discussion.

8 Conclusion

This chapter contains a summarization of the purpose and the research questions. To what extent has the aim been achieved, and what are the answers to the research questions?

The consequences for the target audience (and possibly for researchers and practitioners) must also be described. There should be a section on future work where ideas for continued work are described. If the conclusion chapter contains such a section, the ideas described therein must be concrete and well thought through.

Bibliography

Bessembinder, Hendrik (2018). "Do Stocks Outperform Treasury Bills?" In: *Journal of Financial Economics*.

Campbell, John Y., Andrew W. Lo, and A. Craig MacKinlay (1997). *The Econometrics of Financial Markets*. Princeton University Press.

Cont, Rama (2001). "Empirical properties of asset returns: stylized facts and statistical issues". In: *Quantitative Finance* 1.2, pp. 223–236. DOI: 10.1080/14697680108916295.

Doane, David P. and Lori E. Seward (2011). "Measuring Skewness: A Forgotten Statistic?" In: *Journal of Statistics Education* 19.2. DOI: 10.1080/10691898.2011.11889611.

Financial Edge Training (2022). *Survivorship Bias*. https://www.fe.training/free-resources/asset-management/survivorship-bias/. Accessed: 2026-02-24.

Miskolczi, Panna (2017). "Note on Simple and Logarithmic Return". In: *Applied Studies in Agribusiness and Commerce – APSTRACT* 11.1–2, pp. 127–136. ISSN: 1789-7874. DOI: 10.19041/APSTRACT/2017/1-2/16.

Mota, Pedro P. (2012). "Normality assumption for the log-return of the stock prices". In: *Discussiones Mathematicae: Probability and Statistics* 32, pp. 47–58. DOI: 10.7151/dmps.1143.

Newey, Whitney K. and Kenneth D. West (1987). "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix". In: *Econometrica* 55.3, pp. 703–708.

Reinholtz, Nicholas, Philip M. Fernbach, and Bart de Langhe (2016). *Do People Understand the Benefit of Diversification?* Working Paper. Available at SSRN: 2719144.

Walker, Johnny (1999). "On Whiskey". In: *Communications of the AA*.